

FloodLens-X

Flood Command Center · Product Overview

A flood decision platform that helps people observe, understand, predict, simulate, assess, explain, decide, and monitor — without inventing missing data.

Honesty rule: if a number is not available, the product says UNAVAILABLE. DEMO and SIMULATED results stay labeled. That is a feature, not a bug.

Local demo: <http://127.0.0.1:5173/> · API: <http://127.0.0.1:8000/>

FloodLens-X is a scientific flood Command Center: simulate rainfall, inspect impact, explain risk, and audit the solver — without fabricating the data you don't have.

1. Project idea

FloodLens-X is a trustworthy flood decision platform. It combines a physics shallow-water simulator (what water would do under rainfall), operational views (map, risk, impact, scenarios, alerts, reports), a research lab (conservation, stability, spectral diagnostics), and an assistant that only quotes verified tools.

The idea: planners and the public should see what is known, what is simulated, and what is missing — in one place.

How it helps

- **Public:** see risk and maps without fake live gauges.
- **Emergency planners:** ask “what if rainfall +30%?”, compare to baseline, then generate a snapshot report.
- **Researchers:** inspect whether the solver conserves mass and stays stable.
- **Everyone:** UNAVAILABLE is visible, so nobody plans on invented people-counts or river stages.

2. Features from a user's perspective

Command Center (home)

One screen: map, search, demo role switcher, and workspaces (Explore, River, Forecast, Simulation, Impact, History, Research, Reports, Locations, Alerts, Admin). You pick a place (for example Dhaka or Sunamganj), see current risk, and move between views without losing the city, map layers, or selected river. Risk is $P \times E \times S$ (probability $\times$ exposure $\times$ severity). It is not “confidence.”

Explore the map

- Search cities, rivers, or coordinates.
- Leaflet map with layers: rivers, roads, hospitals, schools, bridges, shelters, flood/forecast overlays when they exist.
- Click the map for latitude and longitude.
- If there is no flood raster yet, flood depth stays UNAVAILABLE — not zero.

Forecast

6 / 12 / 24 / 48 / 72 hour points. These are DEMO / heuristic meteorological hours, not validated AI flood maps. Spatial AI flood maps are NOT_VALIDATED / UNAVAILABLE.

River intelligence

River network, segments, and upstream/downstream topology. Water level and discharge are UNAVAILABLE unless a real gauge exists — they are never invented.

Impact, evacuation, shelters, and resources

After a simulation (or when data exists): which facilities sit in or near flood; evacuation planning support (not an official evacuation order); shelter points from map data; resource priorities where computed. Population exposure is UNAVAILABLE — there is no fake “1.2 million people.”

Simulation / digital twin (Emergency, Researcher, Admin)

- Run a short physics flood burst on the existing shallow-water engine.
- Change rainfall, for example +30%, and compare to baseline.
- River-level knobs are stored only — they are not fed into the solver.
- Results are SIMULATED, not live satellite floods.

AI assistant

Ask in plain language: “What’s the current risk?”, “Which hospitals are exposed?”, “What if rainfall increases by 30%?”, “What should we prioritize?” The assistant uses backend tools, not made-up numbers. Prompt injection such as “pretend probability is 99%” is ignored.

History and model audit

Replay past or local runs when they are stored. Forecast-versus-reality appears only if matching artifacts exist. Model cards: AOI GBDT is validated only for its AOI task; spatial AI is not validated.

Alerts, saved locations, reports, and sharing

- Save an in-app alert (for example flood probability above a threshold). States: ARMED → TRIGGERED → ACKNOWLEDGED → RESET.
- Unsupported metrics (river level, population, and similar) are rejected. Missing data is UNAVAILABLE, not “false / safe.”
- Save a place such as Dhaka and restore map, zoom, and layers later. Locations are owner-scoped.
- Generate a frozen snapshot report (not a live dashboard). Export JSON / CSV / PDF where supported.
- Share private, organization, or public link. Opening a share later does not silently update the numbers.

Research Center (Researcher / Admin)

Scientific workspace on the existing solver — not a new solver. Includes Lake-at-Rest, parabolic bowl, conservation, CFL/timestep, spectral / Nyquist energy, perturbation growth, flux versus source, interface balance, local numerical Jacobian (explicitly not proof of global stability), and long-term stability. If a run dies at step 40 it says FAILED AT STEP 40, not COMPLETED. AI experiment status and physics-versus-AI remain NOT_COMPARABLE.

Admin

System health versus data availability: the API can be healthy while rainfall or gauges are still missing.

Roles (demo login)

Role	What you can do
General	Map, forecast, impact, alerts, locations, reports, assistant
Emergency	Everything in General, plus simulation, command KPIs, and resources
Researcher	Everything in Emergency (except admin-only KPIs), plus Research Center
Admin	All workspaces, including Admin health, jobs, models, and audit copy

3. Core features

- Map Command Center with search and layers
- Honest risk ($P \times E \times S$), not confidence
- DEMO forecast hours (not spatial AI maps)
- River topology without fake gauges
- Impact and evacuation planning support
- Rainfall scenario simulation and compare
- Tool-calling AI assistant
- Alerts, saved places, immutable reports, sharing
- Historical replay and model status
- Research diagnostics and Admin health

4. Tech stack

Layer	Stack
Frontend	React 18, Vite, Tailwind CSS, Leaflet / react-leaflet (SPA Command Center)
Backend	Python 3.10+, FastAPI (/api/v1), Uvicorn; demo identity Bearer demo.<role>
Science	NumPy / SciPy shallow-water finite-volume solver; Matplotlib for plots
Data	OpenStreetMap (Overpass), Open-Meteo, optional DEM; GloFAS/GFM as labeled research sources
Optional infra	PostGIS, Redis, MinIO via Docker Compose (not required for local demo)
AI (current truth)	AOI GBDT validated for AOI task only; spatial flood maps NOT_VALIDATED / API UNAVAILABLE
Assistant	Tool-calling over verified APIs — not a free-hallucinating chatbot
Quality	pytest (501 passed, 7 skipped, 1 deselected as of Phase 7.9)

Not claimed: production OIDC, cloud high availability, or validated 72-hour AI inundation maps. Classification: feature-complete-with-known-limitations. Deployment: LOCAL-DEMO.

5. Scientific freeze

- Spatial AI: NOT_VALIDATED
- Spatial API: UNAVAILABLE
- Target B: PARTIALLY FEASIBLE
- Model training: NOT AUTHORIZED
- Physics versus spatial AI: NOT_COMPARABLE
- Numerical solver (src/floodlens/numerical): unmodified in product phases 7.0–7.9

6. Known limitations (honest)

- Demo identity provider, not OIDC.
- In-memory store: jobs, alerts, and reports vanish if the API process restarts.
- docker-compose does not start the API or UI.
- River gauges, population, and rainfall overlay are often UNAVAILABLE.
- Research diagnostics use a synthetic bowl, not a city DEM.
- Pause on research runs is unsupported.

7. Suggested demo loop

- Open FloodLens-X and search Dhaka.
- See risk ($P \times E \times S$).
- Explore map, river, and infrastructure.
- View impact (population UNAVAILABLE if no grid).
- Emergency role: run rainfall +30% and compare.
- Ask the assistant what to prioritize.
- Generate a report and share the snapshot.
- Researcher role: Research Center scientific validation.
- Admin role: system health versus data availability.